

# Controlled 3D Ground-Penetrating Radar Identifiability of Close Rebars

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## Executive Summary

This report summarizes the current three-dimensional ground-penetrating-radar simulation campaign for close, finite-length steel rebars in a concrete-like medium. The study uses synthetic scenes with known geometry so that each comparison can be interpreted against a known truth.

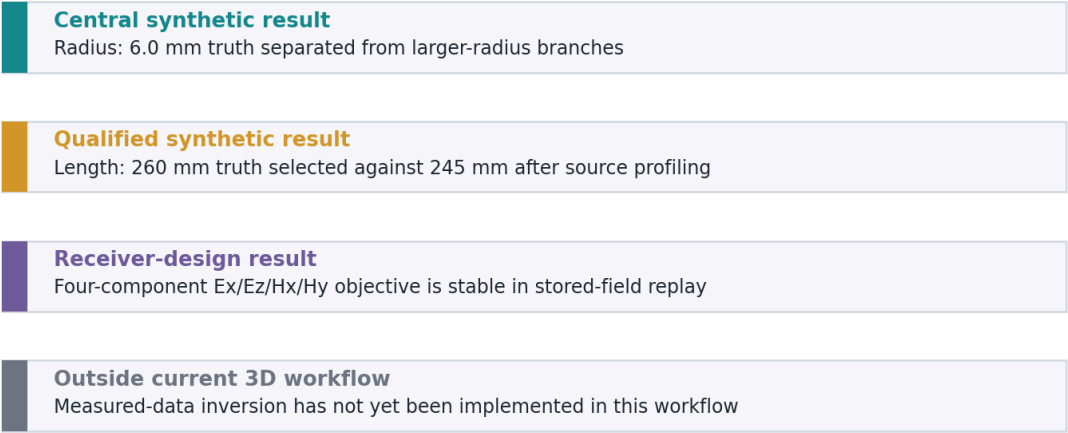
The central result at this stage is radius identifiability in the controlled 1 mm-grid setting. The 6.0 mm radius truth is separated from nearby larger-radius branches on the central fixed-y scan line and on two adjacent fixed-y scan lines under the conservative  $E_y$  and  $H_z$  receiver components. The smallest waveform-tested radius increase currently supported across those lines and components is 0.125 mm.

The local length result is visible but more sensitive. The 260 mm truth is selected against the nearby 245 mm branch after source profiling on four crossline positions, but the same evidence shows that source mismatch can make a wrong length branch look competitive. The length result is therefore treated as a qualified synthetic result rather than as a measured-field length-resolution claim.

Receiver-objective analysis has identified a promising four-component design candidate using  $E_x$ ,  $E_z$ ,  $H_x$ , and  $H_y$ . That finding remains a receiver-design result, not a primary geometry claim, because physical receiver calibration has not yet been integrated into this 3D workflow. Measured-data inversion has also not yet been implemented in this 3D workflow while the synthetic geometry hierarchy and receiver model are being established.

### Current evidence hierarchy

The figure separates supported geometry evidence, qualified geometry evidence, receiver-design evidence, and field-data scope.



**Figure 1. Current evidence hierarchy.** The figure separates the supported geometry result, the qualified length result, the receiver-design result, and the measured-data boundary. Radius response is the central supported result in the current controlled synthetic study. Local length is meaningful but more sensitive to source treatment. The four-component receiver objective is scientifically useful, but it remains below the primary-claim level until physical receiver calibration is integrated.

## Scientific Objective

The purpose of this work is not general rebar detection. The question is narrower and more testable:

In a controlled three-dimensional close-rebar scene with known truth, which nearby wrong geometries can be rejected, and which uncertainty sources can still change that decision?

**Table 1. Evidence layers used in the evaluation.** The table separates geometry evidence from receiver-design evidence and measured-data evidence. This distinction is important because a result can be valid inside a controlled synthetic study without yet being a measured-data inversion result.

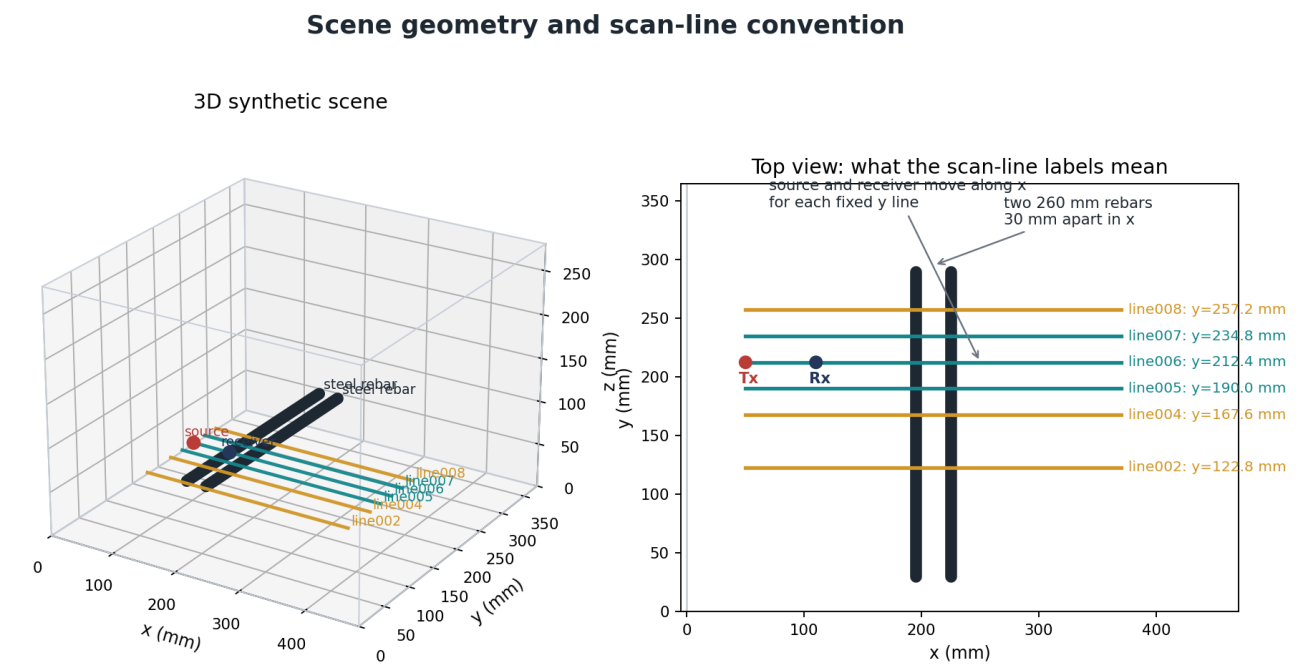
| Evidence layer            | Current role     | Interpretation                                                                                                           |
|---------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------|
| Synthetic radius response | Central result   | Radius is the clearest supported geometry parameter in the tested 1 mm setup.                                            |
| Local length response     | Qualified result | Local length changes are visible in clean simulations but remain sensitive to source and receiver assumptions.           |
| Receiver-design replay    | Design result    | The four-component Ex/Ez/Hx/Hy objective improves stability in stored-field replay but still needs physical calibration. |

| Evidence layer          | Current role                            | Interpretation                                                                                                                            |
|-------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Measured-data inversion | Not yet implemented in this 3D workflow | Field-data inversion is outside the present result set while the 3D synthetic geometry behavior and receiver model are being established. |

## Geometry And Scan-Line Convention

The labels `line002` , `line004` , and similar names are scan-line indices, not physical units. Each label identifies a fixed crossline y position. Along each fixed-y line, the source and receiver move through 35 x positions, producing one B-scan matrix for that line and receiver component.

The close-rebar scene contains two 260 mm steel rebars oriented in the y direction. Their x midpoints are 195 mm and 225 mm, so the pair is separated by 30 mm in x. The source and receiver are above the scene near z = 34 mm; the rebars are centered near z = 90 mm.



**Figure 2. 3D scene and scan-line convention.** The left panel shows the synthetic scene in 3D. The right panel shows the same setup from above. A scan-line label such as `line006` means a fixed y position, here y = 212.4 mm. The transmitter and receiver then move along x for that fixed y position, creating the waveform matrix compared in the inversion tests.

**Table 2. Scan-line labels used in this report.** The table translates the scan-line labels into physical crossline positions and states how each line is used in the analysis.

| Scan-line label | Crossline y position | Role in this report                                            |
|-----------------|----------------------|----------------------------------------------------------------|
| line002         | 122.8 mm             | Dense fixed-y length comparison                                |
| line004         | 167.6 mm             | Dense fixed-y length comparison                                |
| line005         | 190.0 mm             | Adjacent radius and receiver-stress comparison                 |
| line006         | 212.4 mm             | Central line for radius, length, material, and receiver checks |
| line007         | 234.8 mm             | Adjacent radius and receiver-stress comparison                 |
| line008         | 257.2 mm             | Dense fixed-y length comparison                                |

A source-position index is the x-location number within a scan line. For example, source position 1 and source position 35 are two different x locations on the same fixed-y scan line.

## Waveform Mismatch Definition

Each geometry candidate produces a waveform matrix. Rows correspond to source positions, and columns correspond to time samples. The comparison uses a weighted relative L2 mismatch after allowing small nuisance changes in amplitude and time shift:

$$m(S,O)=\min_{a,\tau}\frac{\left\|W\left(aS_{\{\tau\}}-O\right)\right\|_2}{\left\|WO\right\|_2}.$$

Here,  $O$  is the observed waveform matrix from the known-truth simulation,  $S$  is the candidate synthetic waveform matrix,  $W$  selects or weights the time gate,  $a$  is a fitted amplitude scale, and  $\tau$  is a small time shift. A mismatch of zero would mean the gated candidate waveform is identical to the gated observation after the allowed nuisance fit. A mismatch of 0.08 means the remaining residual norm is about 8 percent of the gated observed waveform norm.

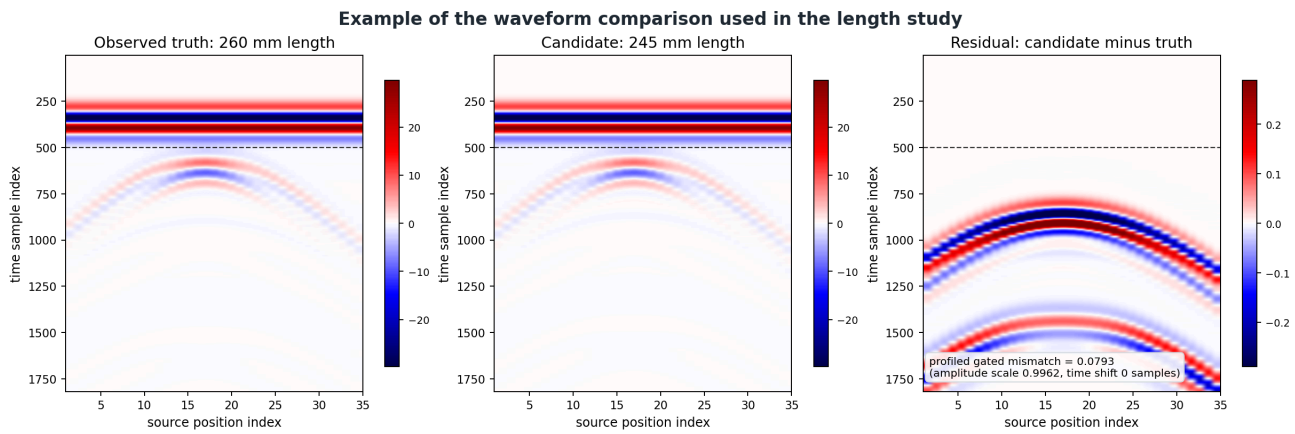
When source frequency is profiled, the geometry score is the best mismatch over the tested source choices:

$$M(g)=\min_{f\in F}m(S_g,f,O).$$

The margin used in the ranking tables is:

$$\Delta=M(\text{nearest wrong geometry})-M(\text{truth geometry}).$$

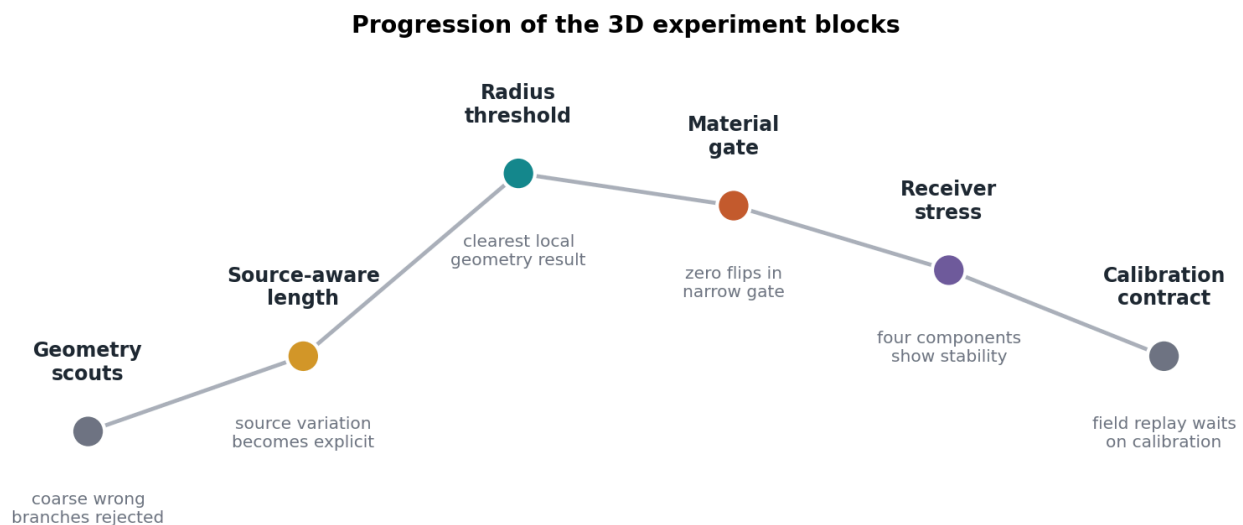
A larger positive margin means the nearest wrong geometry is easier to reject. In the synthetic tables, the truth geometry is known, so the margin measures how far the closest wrong branch remains from the truth after the same nuisance treatment.



**Figure 3. Example waveform mismatch comparison.** This figure shows the actual waveform object being compared in the length study. The left panel is the Ey B-scan for the known 260 mm truth, and the middle panel is the Ey B-scan for a 245 mm length candidate on `line005`. Each column position corresponds to one source location along the fixed-y scan line, and each vertical position is a time sample. The dashed horizontal line marks the start of the time gate used in the reported mismatch, so later samples carry the comparison. The right panel is the residual, candidate minus truth; this residual is what remains after the allowed amplitude and time-shift adjustment and is the quantity normalized in the mismatch equation.

## Evolution Of The Experiment Blocks

The scientific position changed in stages. Early close-pair tests rejected coarse wrong explanations. The dense fixed-y length study then showed that local length is visible only when source variation is handled carefully. The later 1 mm radius-threshold work produced a clearer geometry result. Material perturbation and receiver replay then defined the current limits of that result.



**Figure 4. Progression of the experiment blocks.** Each point represents a result-driven stage of the campaign. The progression moved from geometry scouts, to source-aware length testing, to radius-threshold testing, to material and receiver checks. The final stage shown here is the receiver-calibration contract, which keeps multi-component receiver evidence bounded until physical calibration is integrated.

**Table 3. How the evidence changed the scientific position.** The entries are organized by result, not by file path or internal storage location.

| Stage                             | Result                                                                                                                         | Effect on the scientific position                                          |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Close-pair geometry scouts        | Coarse wrong branches were rejected.                                                                                           | The campaign moved from feasibility into local ambiguity testing.          |
| Dense fixed-y length study        | The 260 mm truth was selected against the 245 mm branch on four crossline positions after source profiling.                    | Local length is visible, but source treatment is essential.                |
| 1 mm radius-threshold study       | A 0.125 mm radius increase was resolved under Ey and Hz on line005 , line006 , and line007 .                                   | Radius became the central supported local geometry result.                 |
| Narrow material perturbation gate | Four homogeneous material perturbations produced zero geometry-ranking reversals across 32 checks.                             | The tested material offsets did not overturn the geometry ranking.         |
| Receiver-objective replay         | The four-component Ex/Ez/Hx/Hy objective had zero ranking reversals in the available three-line stress evidence.               | The receiver design became promising, but still not physically calibrated. |
| Receiver-calibration contract     | 12 required calibration items are prepared as intake categories, with zero real items integrated into this 3D workflow so far. | Physical receiver replay remains a later stage of the work.                |

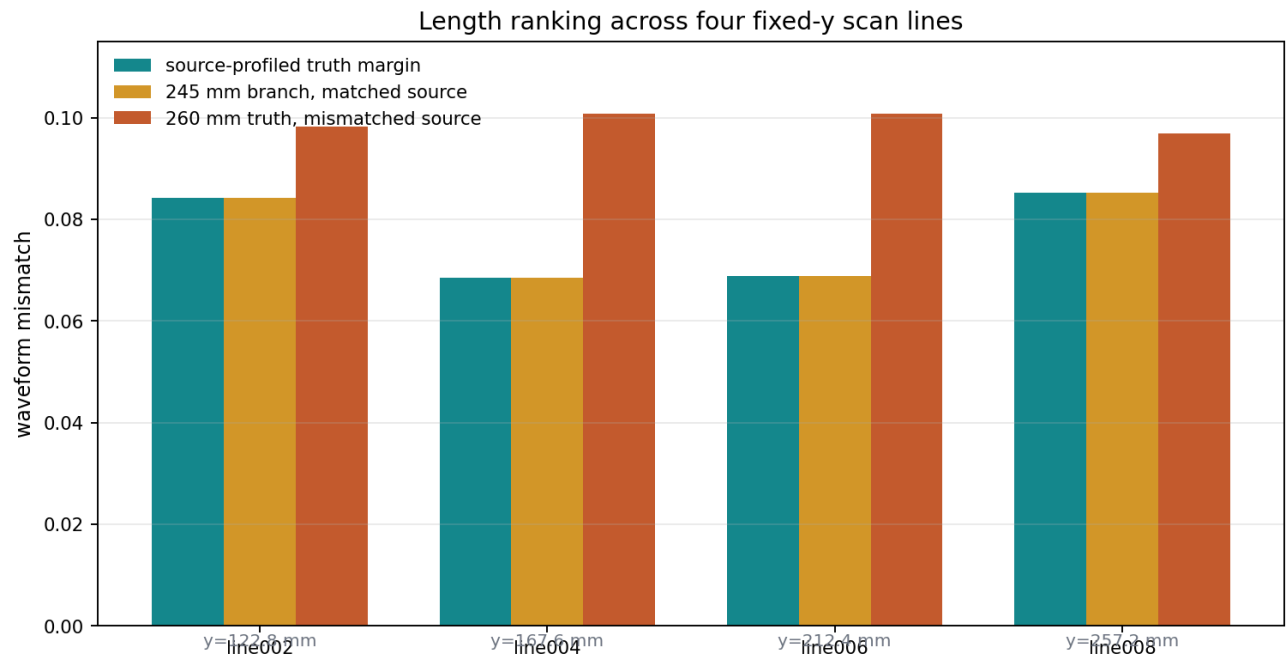
## Result 1: Length Is Visible, But Source Treatment Is Essential

The dense fixed-y study compared the 260 mm truth with the nearby 245 mm length competitor on line002 , line004 , line006 , and line008 . Those labels correspond to  $y = 122.8$  mm, 167.6 mm, 212.4 mm, and 257.2 mm, respectively. Each geometry was evaluated under two source frequencies. After source profiling, all four lines selected the 260 mm truth.

**Table 4. Dense fixed-y length ranking.** The truth margin is the separation between the selected 260 mm geometry and the nearest wrong geometry after source profiling. The last two columns show the source-confounding risk: a wrong geometry with the matched source can look better than the truth evaluated with a mismatched source.

| Scan line | Crossline y position | Truth margin | 245 mm branch with matched source | 260 mm truth with mismatched source |
|-----------|----------------------|--------------|-----------------------------------|-------------------------------------|
| line002   | 122.8 mm             | 0.084182     | 0.084182                          | 0.098316                            |
| line004   | 167.6 mm             | 0.068551     | 0.068551                          | 0.100825                            |
| line006   | 212.4 mm             | 0.068920     | 0.068920                          | 0.100758                            |
| line008   | 257.2 mm             | 0.085154     | 0.085154                          | 0.096952                            |

The length evidence is meaningful because the truth remains selected after source profiling. It is also limited because a fixed-source comparison would be misleading. The result is best read as controlled synthetic length sensitivity under explicit source treatment.



**Figure 5. Source-aware length comparison across four fixed-y scan lines.** Blue bars show the source-profiled margin separating the 260 mm truth from the 245 mm competitor. Gold bars show the 245 mm competitor with the matched source. Orange bars show the true geometry with a mismatched source. The orange bars being larger than the gold bars show why source uncertainty must be part of the interpretation.

## Result 2: Radius Is The Clearest Geometry Parameter

The radius study tests whether the simulation can detect small radius changes in the same close-pair geometry. The current evidence supports the 6.0 mm radius truth against larger-radius branches on the central scan line and two adjacent scan lines.

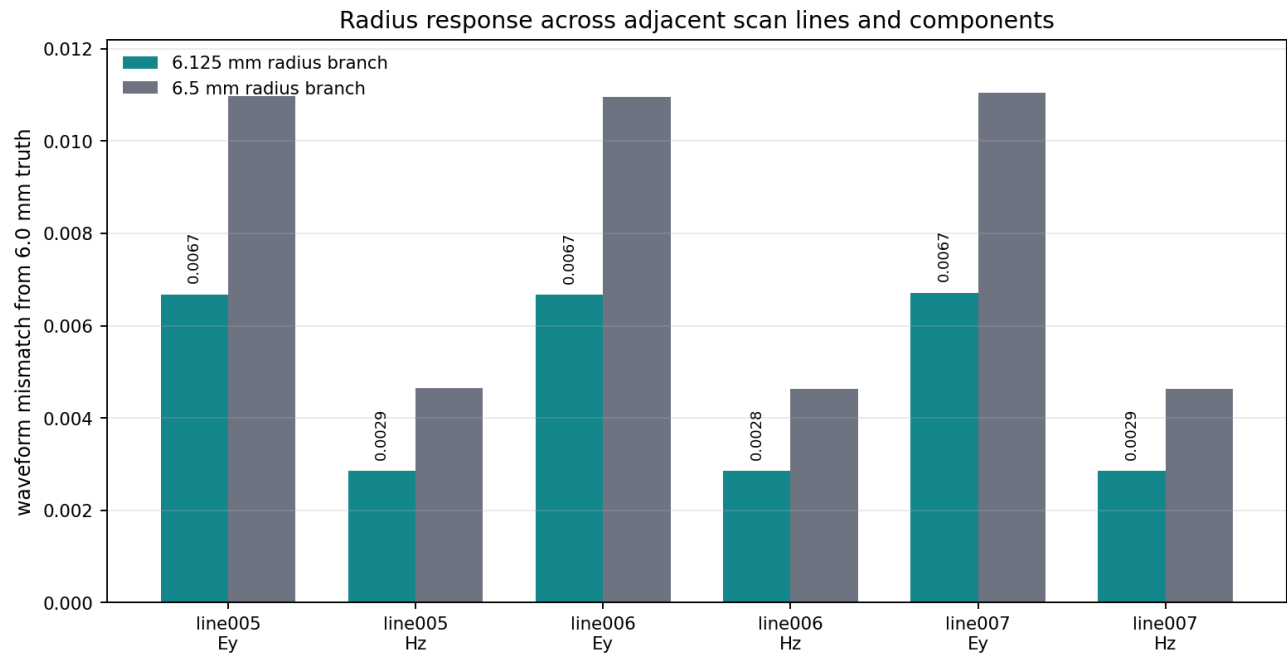
The smallest waveform-tested radius increase currently supported across all six reported checks is 0.125 mm. That means the 6.125 mm branch is separated from the 6.0 mm truth

under Ey and Hz on line005 , line006 , and line007 . The larger 6.5 mm branch is also separated consistently.

**Table 5. Radius response threshold.** The table reports waveform mismatch from the 6.0 mm truth for the 6.125 mm branch and for the larger 6.5 mm branch. All listed entries are nonzero in the current source-profiled evidence.

| Scan line | Crossline y position | Component | 6.125 mm branch mismatch | 6.5 mm branch mismatch |
|-----------|----------------------|-----------|--------------------------|------------------------|
| line005   | 190.0 mm             | Ey        | 0.006675                 | 0.010973               |
| line005   | 190.0 mm             | Hz        | 0.002855                 | 0.004639               |
| line006   | 212.4 mm             | Ey        | 0.006670                 | 0.010964               |
| line006   | 212.4 mm             | Hz        | 0.002847                 | 0.004626               |
| line007   | 234.8 mm             | Ey        | 0.006714                 | 0.011043               |
| line007   | 234.8 mm             | Hz        | 0.002852                 | 0.004638               |

A separate geometry-rendering check adds an important boundary: below the waveform-tested threshold, some very small radius changes can render identically on the 1 mm grid. Therefore, the supported waveform claim is the 0.125 mm result, not arbitrary sub-grid radius recovery.



**Figure 6. Radius response in the tested 1 mm synthetic setup.** The teal bars show the mismatch for the 6.125 mm branch, and the gray bars show the larger 6.5 mm branch. The consistent nonzero mismatch across all six checks is why radius is now the clearest local geometry result.



# Result 3: Radius Is Less Fragile Than The Local Length Signal

The radius and local length results are not equally robust. In the current evidence, the radius separation is about 6.31 times larger than the local length separation in the clean margin comparison. Under the structured error threshold comparison, the ratio is about 6.25.

A clean margin is the nearest wrong mismatch minus the truth mismatch:

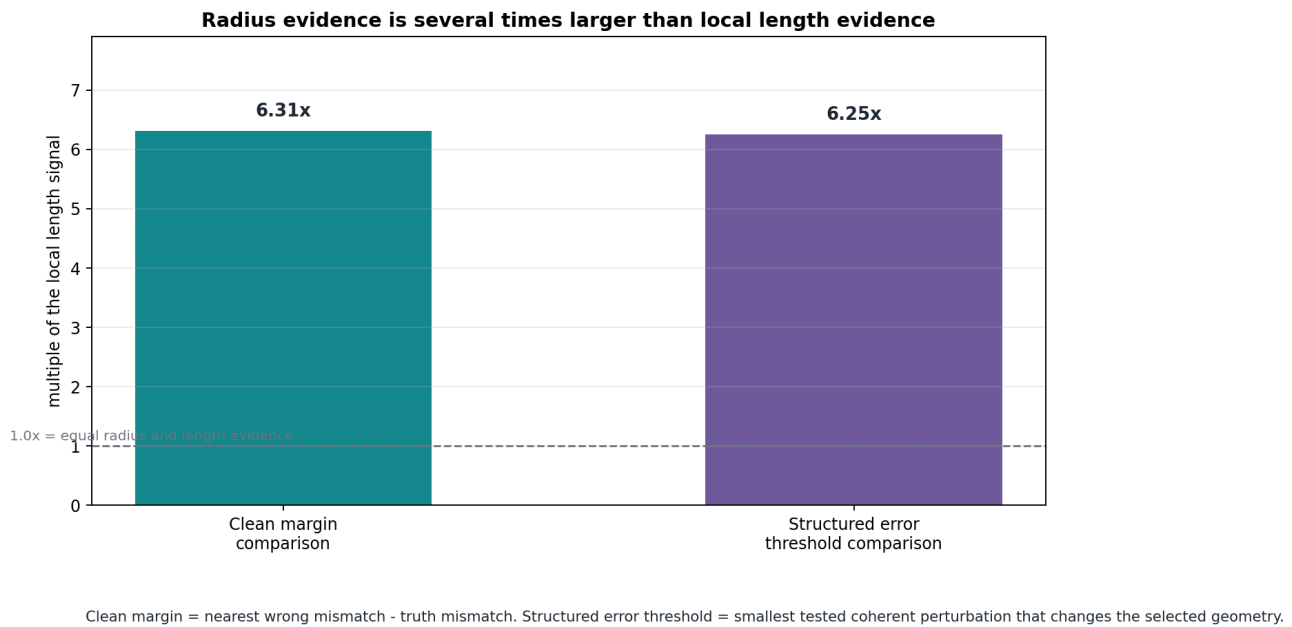
$$C = M(\text{nearest wrong geometry}) - M(\text{truth geometry}).$$

A structured error threshold is the smallest tested coherent perturbation that changes the selected geometry. Here, structured error means a physically organized change, such as source mismatch, material offset, or receiver residual stress, rather than independent random noise added sample by sample.

**Table 6. Relative robustness of radius and local length.** The values divide the radius evidence by the local length evidence under the same style of comparison. A value of 6.31x means the radius separation is about 6.31 times the local length separation for that comparison.

| Comparison                            | Definition                                                              | Radius-to-length ratio | Interpretation                                                           |
|---------------------------------------|-------------------------------------------------------------------------|------------------------|--------------------------------------------------------------------------|
| Clean margin comparison               | Nearest wrong mismatch minus truth mismatch                             | 6.31x                  | Radius has substantially more separation than the local length signal.   |
| Structured error threshold comparison | Smallest tested coherent perturbation that changes the geometry ranking | 6.25x                  | Radius tolerates more structured perturbation before a ranking reversal. |

This is the reason the report treats radius as the central result and length as qualified. The length signal is real in the clean synthetic tests, but it sits closer to the uncertainty boundary.



**Figure 7. Radius-to-length robustness comparison.** The vertical axis reports how many times larger the radius evidence is than the local length evidence under the same comparison type. The dashed 1.0x line is the reference case where radius and length would have equal support. The clean-margin bar compares direct separation from the nearest wrong geometry; the structured-error bar compares how much coherent perturbation is required before the selected geometry changes. Both bars are near 6x, which is why the radius result is treated as the central geometry finding while the local length result remains qualified.

## Result 4: The Narrow Material Gate Did Not Change The Geometry Ranking

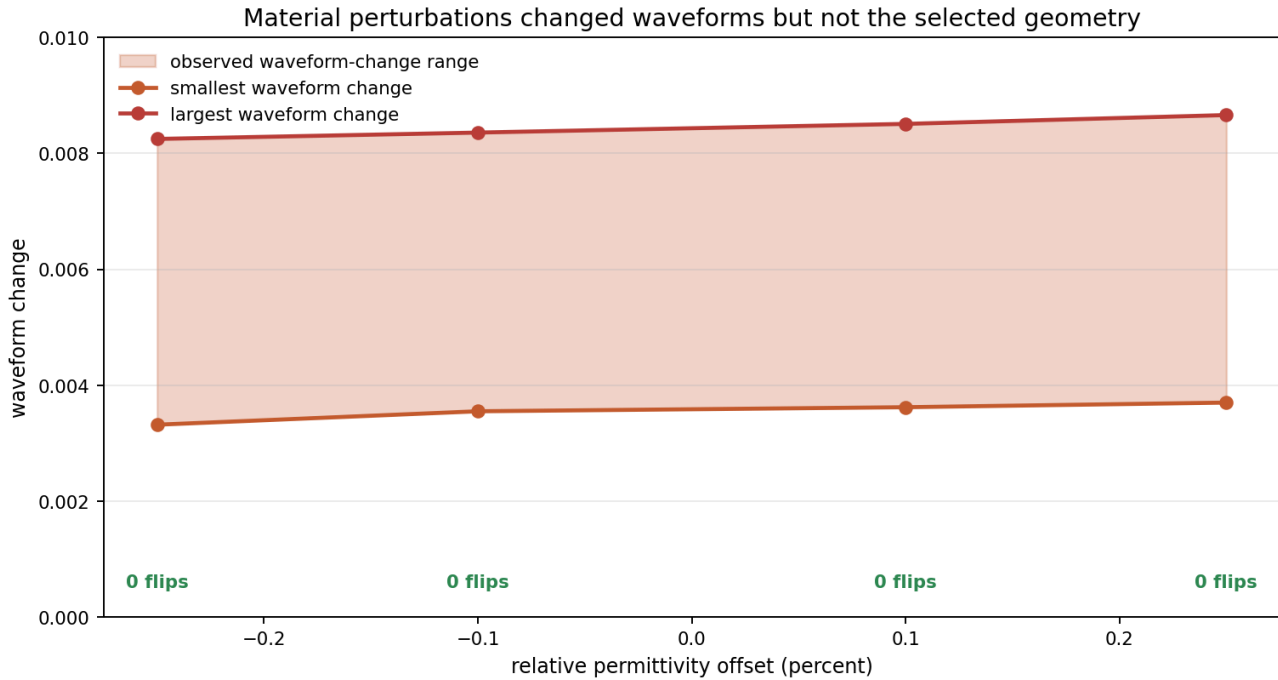
A material perturbation study tested whether small homogeneous changes in relative permittivity could overturn the geometry decision for the `line006` truth case. Four perturbations were evaluated: -0.25 percent, -0.10 percent, +0.10 percent, and +0.25 percent.

**Table 7. Material perturbation gate.** The table reports the scientific outcome rather than job details. The material offsets changed the waveforms, but did not change the selected geometry in this narrow test.

| Quantity                    | Value    |
|-----------------------------|----------|
| Material perturbation cases | 4        |
| Ranking checks              | 32       |
| Geometry-ranking reversals  | 0        |
| Smallest waveform change    | 0.003318 |

| Quantity                | Value    |
|-------------------------|----------|
| Largest waveform change | 0.008662 |

This reduces one uncertainty risk for the tested `line006` case. It does not establish broad material robustness, because the gate is homogeneous, synthetic, and intentionally narrow.



**Figure 8. Material perturbations changed waveforms but did not change the selected geometry.** The horizontal axis shows the relative permittivity offset. The vertical band shows the observed waveform-change range. The selected geometry stayed on the truth branch across the checked perturbations.

## Result 5: The Four-Component Receiver Objective Is Promising But Not Yet Primary

The receiver analysis replayed stored field components to test whether a richer receiver objective could improve stability. The most important design candidate combines  $E_x$ ,  $E_z$ ,  $H_x$ , and  $H_y$ . In this report, it is called the four-component receiver objective.

For a component set  $C$ , the reported multi-component mismatch is summarized as:

$$m_C = \sqrt{\frac{1}{|C|} \sum_{c \in C} m_c^2}.$$

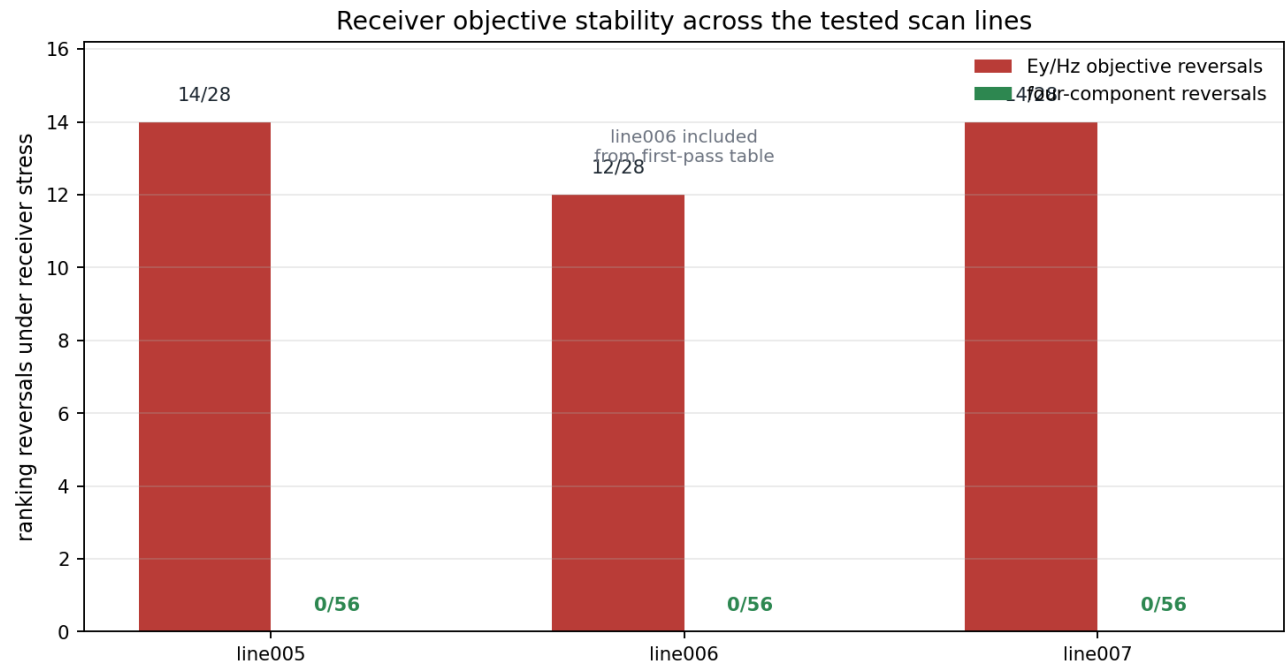
Here,  $m_c$  is the waveform mismatch for one receiver component after the same time gate, amplitude-scale adjustment, and small time-shift adjustment used in the single-component

comparisons. The component set C is the group of receiver components being combined; for the four-component objective,  $C=\{Ex,Ez,Hx,Hy\}$ . The square-root average keeps the combined score on the same general scale as an individual component mismatch, so the score is not larger merely because more channels were included. This objective had zero ranking reversals in the available three-line receiver-stress evidence. The current conservative Ey/Hz objective remains the basis for the supported geometry claims, but it is more fragile under these stress checks.

**Table 8. Receiver-stress comparison.** A ranking reversal means that an imposed receiver residual perturbation changed the selected geometry. line006 is included from an earlier first-pass stress table, while line005 and line007 use the later matched-profile stress table. This is why the line006 row is marked as a protocol difference rather than omitted.

| Scan line | Crossline y position | Ey/Hz reversals | Four-component reversals | Note                            |
|-----------|----------------------|-----------------|--------------------------|---------------------------------|
| line005   | 190.0 mm             | 14 / 28         | 0 / 56                   | matched-profile stress table    |
| line006   | 212.4 mm             | 12 / 28         | 0 / 56                   | earlier first-pass stress table |
| line007   | 234.8 mm             | 14 / 28         | 0 / 56                   | matched-profile stress table    |

This is a receiver-design result. A physical instrument measures voltages, not stored simulation field components directly. Before this objective can become primary evidence, the project needs transfer response, gain, timing, cross-talk, background, and noise-covariance calibration in the 3D workflow.



**Figure 9. Receiver-objective stability across the tested scan lines.** The figure compares two ways of scoring the same stored synthetic receiver data under receiver-stress perturbations. A ranking reversal means that the stress perturbation caused the wrong geometry to rank ahead of the truth geometry. Red bars count reversals for the conservative Ey/Hz objective, which remains the objective used for the primary geometry claims. Green bars count reversals for the four-component Ex/Ez/Hx/Hy objective summarized by the equation above. In this diagnostic replay, the four-component objective has zero observed reversals, meaning every tested perturbation kept the selected geometry on the truth branch. The `line006` row is included because it provides useful central-line evidence, but it comes from an earlier first-pass stress table rather than the later matched-profile protocol used for `line005` and `line007` ; this is why the row is interpreted as consistent receiver-design evidence rather than a fully matched replicate.

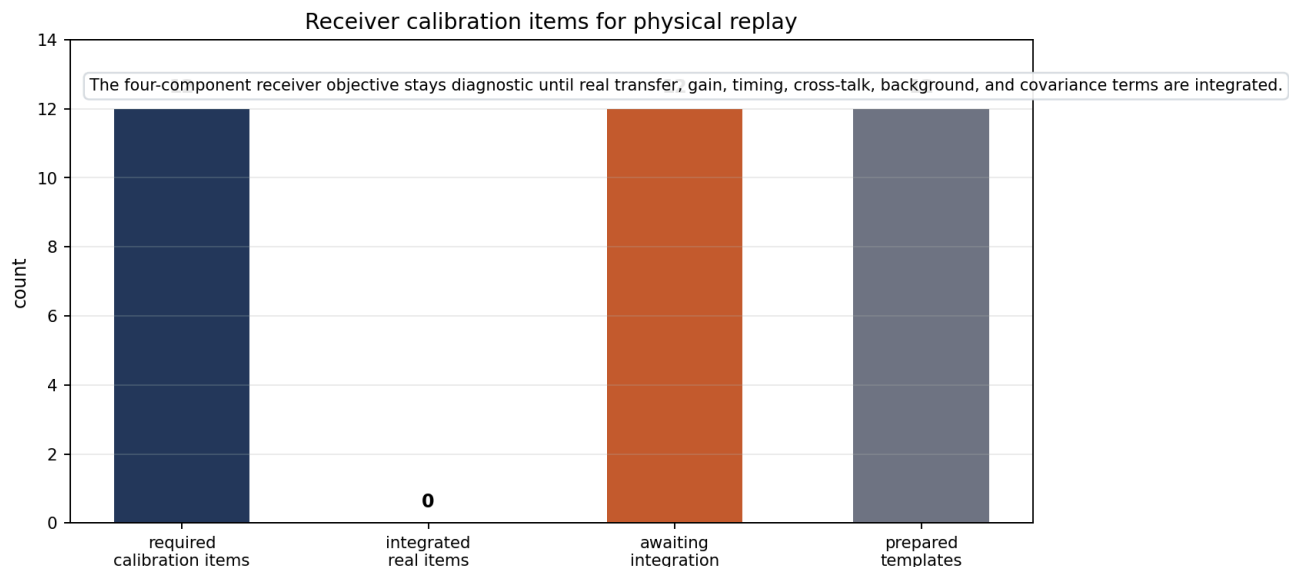
## Result 6: Physical Receiver Calibration Is The Active Limitation

The four-component receiver objective cannot be treated as primary evidence until it is connected to a physical receiver model. The calibration package is not a formality. It defines how simulated field components become predicted measured voltages and how residuals should be weighted.

**Table 9. Receiver-calibration package status.** The required categories include channel ordering and units, receiver observability, transfer response, gain, cross-talk, timing offsets, source filtering, background or bias, noise covariance, calibration provenance, a physical replay manifest, and a final review packet.

| Calibration category                                  | Count |
|-------------------------------------------------------|-------|
| Required calibration categories                       | 12    |
| Integrated real calibration items in this 3D workflow | 0     |
| Categories awaiting integration                       | 12    |
| Prepared intake templates                             | 12    |

This is the main reason the four-component receiver objective remains a design candidate. The stored-field replay is encouraging, but the physical receiver model has not yet been integrated into the 3D workflow.



**Figure 10. Receiver-calibration package status.** The receiver-calibration categories are defined, and intake templates exist for them. Real calibration values still need to be integrated before physical receiver replay can be used as primary geometry evidence.

## Computational Scope

The evidence summarized here comes from completed synthetic simulations and offline replays of completed traces. The current scientific bottleneck is not a larger simulation table. It is the physical receiver model needed to interpret multi-component receiver objectives and the measured-data inversion stage that has not yet been implemented in this 3D workflow.

Additional high-performance simulation is best tied to a specific unresolved ambiguity that cannot be resolved from completed traces. At the present stage, completed traces already support the radius result, the qualified length result, the narrow material gate, and the receiver-design replay. The next scientific leverage comes from receiver calibration and measured-data workflow integration rather than broadening the synthetic grid without a targeted question.

## Current Scientific Position

The current study supports the following position:

Controlled 3D synthetic identifiability of close finite-length rebar geometry, with radius as the central local result, length as a qualified local result, and receiver-objective promotion dependent on physical calibration.

The position is intentionally narrow. It does not claim measured-field performance, arbitrary rebar arrangements, broad material heterogeneity, or completed measured-data full

waveform inversion. It states that, in the tested ideal 1 mm synthetic setup, radius differences are resolved more consistently than local length differences, while receiver design has a promising but still uncalibrated path.

## Scientific Plan

The plan follows from the current evidence hierarchy:

1. The radius-response result will remain the central geometry result for the controlled synthetic study.
2. The local length result will continue to be interpreted through source profiling, material control, and receiver assumptions rather than as a measured-field length-resolution claim.
3. The narrow material-gate result will be used as a bounded uncertainty check, not as a broad material-robustness proof.
4. The four-component receiver objective will remain a receiver-design candidate until a physical transfer response, gain model, timing model, cross-talk model, background term, and noise covariance are integrated.
5. Measured-data inversion will enter the 3D workflow after the synthetic geometry hierarchy and physical receiver model are sufficiently established.
6. Any new simulation work will be tied to a specific ambiguity that is not already answerable from completed traces.

## Conclusion

The three-dimensional campaign now has a clear scientific hierarchy. Radius is the most defensible local geometry result in the current synthetic setup. The evidence resolves a 0.125 mm radius increase under  $E_y$  and  $H_z$  across three tested scan lines and also separates the larger 6.5 mm branch from the 6.0 mm truth.

The local length result remains important, but it is more sensitive. The 260 mm truth is recovered against the 245 mm branch after source profiling, while the same data show why source mismatch can make a wrong geometry appear competitive. The material gate reduces one uncertainty concern by showing zero geometry-ranking reversals for small homogeneous permittivity perturbations, but it does not convert local length discrimination into a measured-field claim.

The receiver-design analysis is also important. A four-component  $E_x/E_z/H_x/H_y$  objective is stable in the available synthetic stress evidence. Its role is still diagnostic until physical receiver calibration is integrated. The current evidence therefore supports a focused result: controlled synthetic 3D identifiability of close rebar geometry, led by radius response, with local length and receiver design treated within their current limits.